

Automated Bifurcation Analysis of Two-Parameter Families of Two-Dimensional Dynamical Systems

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Abstract

Dynamical systems govern a wide range of real-world phenomena, from cardiac rhythms and ecological populations to engineering structures and climate systems. An important challenge in analyzing these systems is identifying bifurcations, which are critical parameter values where a system's qualitative behavior changes fundamentally, potentially causing unexpected outcomes. Detecting these points manually across multiple parameter values is tedious and prone to mistakes, requiring the need for tools that can automate this process. This project presents an automated computational tool for detecting and visualizing bifurcations in two-parameter families of two-dimensional first-order dynamical systems.

1 Mathematical Background

Here we introduce the mathematical concepts that the tool uses to find bifurcations. We begin by defining equilibria and stability, then introduce the Jacobian matrix and eigenvalue analysis as tools for classifying equilibrium behavior, and finally define the two types of bifurcations the tool is designed to detect, Hopf and saddle-node bifurcations.

1.1 Equilibria

Consider a two-dimensional first-order dynamical system of the form:

$$\frac{dx}{dt} = f(x, y, \alpha, \beta) \tag{1}$$

$$\frac{dy}{dt} = g(x, y, \alpha, \beta) \tag{2}$$

where x and y are dependent variables and α, β are parameters. An equilibrium is a point (x^*, y^*) where the system is at rest, meaning that both rates of change are zero simultaneously:

$$f(x^*, y^*, \alpha, \beta) = 0 \quad \text{and} \quad g(x^*, y^*, \alpha, \beta) = 0 \tag{3}$$

At an equilibrium point, the system has no tendency to move, so if placed exactly at (x^*, y^*) it will remain there indefinitely.

1.2 Stability

In a dynamical system, finding the equilibria alone is not sufficient. We also need to understand whether nearby trajectories are attracted to or repelled from the equilibria. An equilibrium is called **stable** if trajectories starting close to it converge toward it over time, and **unstable** if they diverge away.

1.3 The Jacobian Matrix

To determine stability analytically, we linearize the system around an equilibrium (x^*, y^*) using the Jacobian matrix:

$$J = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix} \bigg|_{(x^*, y^*)} \quad (4)$$

The Jacobian, evaluated at an equilibrium, gives a linear approximation of the system around the equilibrium. The eigenvalues of that linear approximation accurately reflect the local behavior of the nonlinear system near the equilibrium.

1.4 Eigenvalue Analysis

The stability of an equilibrium is determined by the eigenvalues λ_1, λ_2 of the Jacobian matrix. Since the Jacobian for two dimensional systems is a 2×2 matrix, it has two eigenvalues which may be real or complex. The equilibrium classification given by the eigenvalues is the primary way our tool detects bifurcations. The classification can be seen in the table below.

Eigenvalues	Equilibrium Classification
Both $\text{Re}(\lambda) < 0$	Stable (trajectories converge)
Both $\text{Re}(\lambda) > 0$	Unstable (trajectories diverge)
$\text{Re}(\lambda)$ of opposite sign	Saddle (stable in one direction, unstable in another)
Both $\text{Re}(\lambda) = 0$, $\text{Im}(\lambda) \neq 0$	Potential Bifurcation (Hopf)
$\text{Re}(\lambda) \approx 0$, $\text{Im}(\lambda) \approx 0$	Potential Bifurcation (saddle-node)

Table 1: Equilibrium classification based on Jacobian eigenvalues.

1.5 Bifurcations

A bifurcation occurs when a small change in a parameter causes a qualitative change in the system's behavior. This project detects two types: **Hopf** and **saddle-node bifurcations**. Both the Hopf and the saddle-node bifurcation types are detected when the eigenvalues of the Jacobian, evaluated at the equilibrium point, cross the imaginary axis as the parameters vary.

Hopf bifurcations occur when a pair of complex conjugate eigenvalues crosses the imaginary axis as the parameters vary. The eigenvalues look like $0 \pm \omega i$ at the bifurcation point, with $\omega \neq 0$. Because our Jacobian is a square matrix, its trace equals the sum of its eigenvalues. At a Hopf bifurcation, the trace of the Jacobian must be equal to 0, and the determinant must be positive.

$$\text{tr}(J) = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} = 0 \quad (5)$$

$$\det(J) = \frac{\partial f}{\partial x} \frac{\partial g}{\partial y} - \frac{\partial f}{\partial y} \frac{\partial g}{\partial x} > 0 \quad (6)$$

with nonzero imaginary parts. Note that with no real parts in the eigenvalues, it's already known that the determinant is positive. As the parameters vary, the equilibrium transitions between a stable spiral and an unstable spiral, briefly becoming a center at the bifurcation point.

Saddle-node bifurcations occur when a single purely real eigenvalue crosses the imaginary axis as the parameters vary. Since the eigenvalues are purely real, this just means an eigenvalue hits zero. As the parameters vary, the system transitions from having two equilibria, a stable node and a saddle point, to having no equilibria, with the two equilibria merging at the bifurcation point.

2 Implementation

This section describes the computational steps the tool takes to detect and visualize bifurcations. The tool is implemented in Python, making use of the `sympy`, `numpy`, `scipy`, and `matplotlib` libraries.

2.1 Symbolic Setup

The user defines the system by writing the two equations f and g as `sympy` expressions in terms of the symbolic variables x , y , α , and β . The Jacobian matrix is then computed analytically using symbolic differentiation via `sp.diff`, giving exact partial derivatives. Each entry of the Jacobian is then converted into a fast numerical function using `sp.lambdify`, which translates the symbolic expression into a `numpy`-backed Python function that can be evaluated efficiently at any point.

2.2 Parameter Sweep

The tool sweeps a user-defined grid of parameter combinations (α, β) . The ranges and resolution of the grid are specified by the user in the settings section of the code. For each combination of parameter values, the tool attempts to locate all the equilibria of the system and classify their stability.

2.3 Equilibrium Finding

For each (α, β) pair, the tool numerically locates the equilibria by calling `scipy.optimize.fsolve` with multiple initial conditions drawn from a grid that is defined by the user. Using multiple starting points is essential because `fsolve` is a local root finder. It finds the equilibrium nearest to the starting point and may miss others if only one initial condition is used. Each candidate solution is validated by checking that the residual $|f(x^*, y^*)| + |g(x^*, y^*)|$ is below a strict tolerance, which is also defined by the user, and duplicate solutions are discarded.

2.4 Stability Classification

At each equilibrium (x^*, y^*) , the Jacobian matrix is evaluated and its eigenvalues λ_1, λ_2 are computed using `numpy.linalg.eigvals`. The stability of the equilibrium is then classified based on the real and imaginary parts of the eigenvalues. A Hopf bifurcation is detected when the trace of the Jacobian is approximately zero with nonzero imaginary parts in the eigenvalues, indicating that the real parts of the eigenvalues are passing through zero. A saddle-node bifurcation is detected when an eigenvalue has both real and imaginary parts approximately zero.

2.5 Visualization

The results are presented as a two-panel diagram generated using `matplotlib`. The left panel shows a full stability map in the (α, β) parameter space, where each point is colored according to the stability classification of the equilibrium found at that parameter combination. The right panel shows only the bifurcation points, specifically Hopf and saddle-node bifurcations, isolating the bifurcation curves in parameter space.

3 Results

The tool was tested on two systems, the Hopf normal form and the Bogdanov-Takens system. For each system, the tool swept a grid of parameter combinations, located equilibria numerically, classified their stability using the Jacobian, and produced a two-panel bifurcation diagram.

3.1 System 1: Hopf Normal Form

The first system tested was the Hopf normal form:

$$\frac{dx}{dt} = \alpha x - y - x(x^2 + y^2), \quad \frac{dy}{dt} = x + \alpha y - y(x^2 + y^2) \quad (7)$$

This system has a single equilibrium at the origin $(x^*, y^*) = (0, 0)$ for all parameter values. The Jacobian evaluated at the origin is:

$$J = \begin{pmatrix} \alpha & -1 \\ 1 & \alpha \end{pmatrix} \quad (8)$$

with eigenvalues $\lambda = \alpha \pm i$. The Jacobian trace is $\text{tr}(J) = 2\alpha$, which equals zero at $\alpha = 0$. At this point, the eigenvalues are purely imaginary ($\lambda = \pm i$), satisfying the Hopf bifurcation condition. For $\alpha < 0$ the equilibrium is a stable sink, and for $\alpha > 0$ it is an unstable source. The bifurcation is independent of β , so the Hopf bifurcation curve in (α, β) parameter space is the vertical line $\alpha = 0$.

The tool correctly identified the Hopf bifurcation curve at $\alpha = 0$ across all values of β , consistent with the analytical result. The stability map clearly shows the transition from stable sink (blue) to unstable source (red) as α crosses zero, with the equilibrium briefly becoming a center (purple) at the bifurcation point.

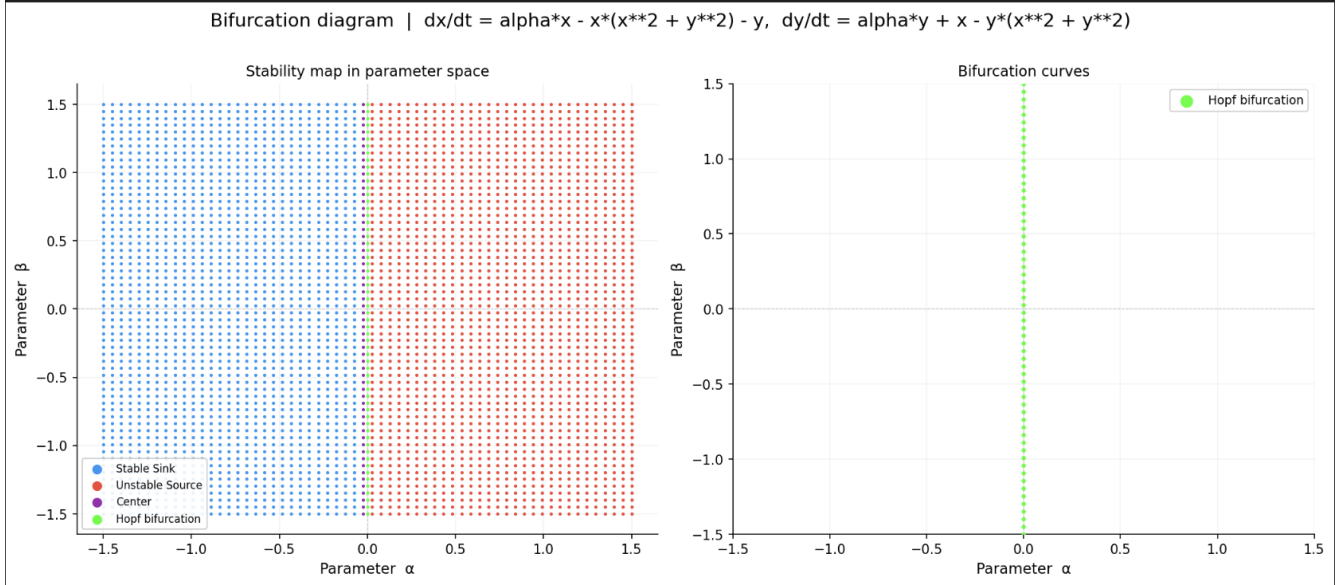


Figure 1: Two-parameter bifurcation diagram for the Hopf normal form. The left panel shows the stability map: blue dots indicate stable sink, red dots indicate unstable source, and purple dots indicate the center at the bifurcation boundary. The right panel isolates the Hopf bifurcation curve, which appears as a vertical line at $\alpha = 0$, consistent with the analytical result.

3.2 System 2: Bogdanov-Takens System

The second system tested was the Bogdanov-Takens normal form:

$$\frac{dx}{dt} = y, \quad \frac{dy}{dt} = \alpha + \beta y + x^2 + xy \quad (9)$$

This system has two equilibria at $(\pm\sqrt{-\alpha}, 0)$ for $\alpha \leq 0$, which collide and annihilate at $\alpha = 0$. Unlike the Hopf normal form, the bifurcation boundary in (α, β) space is a curve rather than a straight line, making it a more demanding test for the tool.

The Hopf bifurcation condition $\text{tr}(J) = 0$ at the equilibrium $(-\sqrt{-\alpha}, 0)$ gives:

$$\beta = \sqrt{-\alpha} \quad (10)$$

which is a curve in the (α, β) space defined for $\alpha \leq 0$ and $\beta \geq 0$. The tool successfully detected the Hopf bifurcation curve, consistent with the calculated curve $\beta = \sqrt{-\alpha}$. The stability map clearly shows the transition from stable sink (blue) to unstable source (red) as the parameters cross the boundary defined by the bifurcation curve. Near this boundary, the equilibrium briefly becomes a center. This is captured by the stability map as purple dots can be seen near the curve.

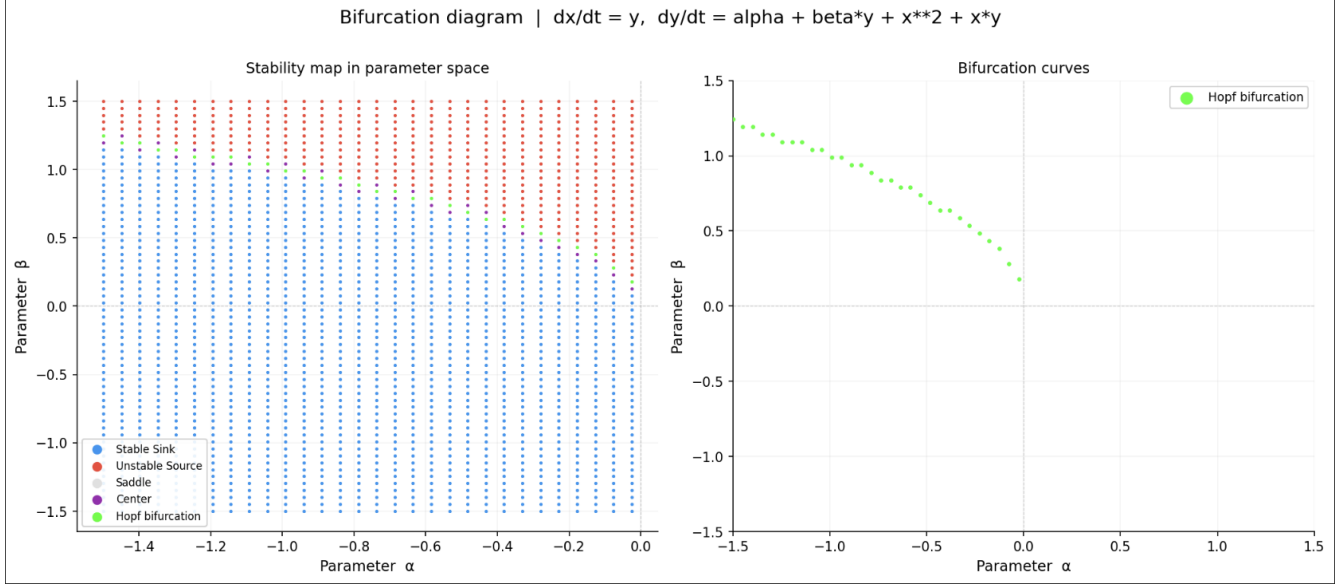


Figure 2: Two-parameter bifurcation diagram for the Bogdanov-Takens system. The left panel shows the stability map across (α, β) parameter space. The right panel isolates the Hopf bifurcation curve, which follows the calculated curve $\beta = \sqrt{-\alpha}$ for $\alpha \leq 0$.

4 Discussion

4.1 Strengths of the Tool

The tool successfully automates the detection and visualization of Hopf bifurcations in two-parameter families of two-dimensional dynamical systems. For both test systems, the tool correctly identified the bifurcation curves and produced clear two-panel diagrams that differentiate between stable and unstable regions in parameter space. A key strength is its generality, the user only needs to define the equations f and g as symbolic expressions. The tool handles the rest automatically, including symbolic differentiation to construct the Jacobian, root finding, eigenvalue analysis, and visualization. This makes it applicable to a broad class of dynamical systems without requiring any manual analysis.

4.2 Limitations

Grid resolution sensitivity. The tool sweeps a finite grid of parameter combinations, meaning bifurcation points that fall between grid points may be missed entirely. This was observed during testing. For the Hopf normal form, the bifurcation at $\alpha = 0$ was initially missed because

the parameter grid never landed exactly on $\alpha = 0$. The fix required either forcing $\alpha = 0$ into the grid explicitly, or using a sufficiently fine grid. This is a fundamental limitation of any grid-based approach.

Saddle-node detection. While the tool is designed to detect saddle-node bifurcations, this was not successfully demonstrated in practice. For the Bogdanov-Takens system, the saddle-node bifurcation at $\alpha = 0$ was not detected. The saddle-node condition requires an eigenvalue to be essentially exactly zero, which only occurs at a precise parameter value. Since the parameter grid is discrete, the grid rarely lands exactly on the bifurcation point, and the eigenvalue at the nearest grid point is not close enough to zero to trigger detection. This is a more fundamental limitation than the Hopf case. The Hopf condition uses the trace, which varies smoothly and gradually approaches zero, making it easier to catch.

Scope. The tool only handles two-dimensional first-order systems with two parameters. Higher-dimensional systems, higher-order systems, and systems with more than two parameters are outside its scope. Additionally, it is designed to detect only Hopf and saddle-node bifurcations. Other types of bifurcations cannot be detected with this tool.